

Student Management System

1. Project Overview

The Student Management System is an interactive web application developed to manage student records efficiently. The project allows users to add, view, search, filter, edit, update, and delete student information. It provides a simple and user-friendly interface for managing details such as student name, email, course, and marks. This project was developed as part of the Week 5 JavaScript Project Assignment.

2. Features Implemented

The project includes several useful features:

- Add new student records.
- Edit and update existing student information.
- Delete student records with confirmation.
- Search students by name.
- Filter students by course and Pass/Fail result.
- Sort students according to marks.
- Calculate total students, average marks, and highest marks.
- Validate form inputs before adding or updating records.
- Dynamically display student records on the webpage.

3. Technologies Used

The project was developed using **HTML5, CSS3, and JavaScript**. HTML5 was used to create the webpage structure, CSS3 was used for styling and responsive layout, and JavaScript was used to implement the application functionality.

4. JavaScript Concepts Used

The project applies JavaScript concepts learned in Weeks 3 and 4, including variables, functions, arrays, objects, conditional statements, loops, array methods, DOM manipulation, and events. Methods such as `push()`, `filter()`, `find()`, `reduce()`, and `sort()` are used for managing student data.

5. Screenshots of the Project

Screenshot 1: Homepage and navigation section

Student Management System

Manage student records easily and efficiently

[Home](#)[Add Student](#)[Student Records](#)[About](#)

Welcome to Student Management System

This application helps you manage student information including names, email addresses, courses, and marks.

Add Student

Enter the student details below to create a new record.

Student Name

Screenshot 2: Add Student form

Add Student

Enter the student details below to create a new record.

Student Name

Email Address

Select Course

Information Technology

Marks

Add StudentClear Form

Screenshot 3: Student records and statistics

Student Records

View and manage all student information.

Showing Students4

Total Students4

Average Marks79.00

Highest Marks94

Spandhana Pass

Email: spandhana@example.com

Course: CSE (AI-ML)

Marks: 85

EditDelete

Priya Fail

Email: priya@example.com

Course: ECE

Marks: 46

EditDelete

Sravanthi Pass

Email: sravanthisrimanthula@gmail.com

Course: CSE

Marks: 94

EditDelete

Anjali Pass

Email: anjali@gmail.com

Course: IT

Marks: 91

EditDelete

Screenshot 4: Search and filter functionality

Search & Filter Students

Search students or filter records according to course and result.

Search by Name

Anjali

Filter by Course

All Courses

Filter by Result

All Results

Sort by Marks

Default Order

Student Records

View and manage all student information.

Showing Students1

Total Students4

Average Marks79.00

Highest Marks94

Anjali

Pass

Email: anjali@gmail.com

Course: IT

Marks: 91

EditDelete

Screenshot 5: Edit student functionality

Anjali

Pass

Email: anjali@gmail.com

Course: IT

Marks: 95

EditDelete

Screenshot 6: Delete student functionality

Student Records

View and manage all student information.

Showing Students3

Total Students3

Average Marks75.33

Highest Marks95

Spandhana

Pass

Email: spandhana@example.com

Course: CSE (AI-ML)

Marks: 85

EditDelete

Priya

Fail

Email: priya@example.com

Course: ECE

Marks: 46

EditDelete

Anjali

Pass

Email: anjali@gmail.com

Course: IT

Marks: 95

EditDelete

6. Challenges Faced

One challenge was connecting the JavaScript data with the HTML elements dynamically. Another challenge was implementing multiple filters and sorting without losing the original student records.

7. Solutions Implemented

These challenges were solved by using DOM methods to create and update elements dynamically and by using JavaScript array methods such as `filter()`, `find()`, and `sort()` to process student records efficiently.

8. What I Learned

Through this project, I learned how JavaScript can be combined with HTML and CSS to create interactive web applications. I also improved my understanding of DOM manipulation, events, functions, arrays, and objects.

9. Future Improvements

In the future, I can add local storage, login functionality, student profile pages, advanced reports, database connectivity, and export options such as PDF or Excel.